CSCE 611 Midterm Fall 2025

Name: _____ Score: ____/125

Part 1: RISC-V ISA

1. Consider the following snippet of RISC-V code:

```
loop: slli  x2,x2,1
      addi  x3,x3,-1
      bne   x3,x0,loop
```

   a. (5 points) What does this code do?

   Shifts the value in register x2 by to the left by a number of bits as given by the contents of the x3 register and sets x3 to 0.

   b. (5 points) How many instructions are executed when this code is executed with register x3 containing the value of 3?

   9

   c. (5 points) Write an equivalent version of this code that performs the same function but after executing less instructions.

   Acceptable:

```
sll   x2,x2,x3
```

   Technically:

```
andi  x3,x3,0x1F
sll   x2,x2,x3
li    x3,0
```

2. (10 points) Write a snippet of RISC-V assembly code (one or more instructions) that stores the value of 9.125 into register x2 as a (32,3)-format fixed-point number; that is, 32 total bits of which 3 bits are fractional (29 whole bits and 3 fractional bits).

```
li    x2,0x49     # 1001.001 = 0x49
```

   or

```
li    x2,73       # 9.125*2^3 = 73
```

Points: _____

Part 2: SystemVerilog

3. (20 points) Identify any problems (i.e. compile errors or bad practices) with the code below:

```
module bar (input logic [1:0] inp, output logic [2:0] outp);

  outp = inp;

endmodule


module foo (input logic [2:0] a,b, output logic [3:0] c);

 bar mybar(.inp(a),.outp(b));

endmodule
```

(1) in module 'foo', the 'mybar' module cannot assign a value to signal 'b' since it is an input
(2) in module 'foo', a 3-bit signal 'a' is being connected to a 2-bit input port 'inp'
(3) in module 'foo', output signal 'c' is not assigned
(4) in module 'bar', the signal assignment is performed outside of an always statement without an "assign"
(5) in module 'bar', a 2-bit signal 'inp' is assigned to a 3-input signal 'outp'

4. (5 points) To what value will the following SystemVerilog code set to signal b when a = 4'b1011?

```
module bit_manipulator (input [3:0] a, output [3:0] b);
  assign b = {4{a[0]}};
endmodule
```

b = 4'b1111

5. (20 points) Write a SystemVerilog module named 'pulse18'. It accepts a clock signal named 'clk', an active-high asynchronous reset signal named 'rst', and a single-bit output named 'p'. When out of reset, the module will pulse the 'p' output (set to 1'b1 for one cycle) every 18 cycles.

```
module pulse18 (input logic clk, input logic rst, output logic p);
   logic [4:0] cnt = 5'd0;
   always_ff @(posedge clk,posedge rst)
        if (rst || cnt == 5'd17) cnt <= 5'd0; else cnt <= cnt+1'd1;
   assign p = cnt == 5'd17;
endmodule
```

Points: _____

6. (10 points) Convert the behavior of the following Verilog module into a truth table:

```
module test_module(input logic [2:0] a, output logic [1:0] y);
    always_comb begin
        y = 2'b11;
        if (a[2]) y = 2'b00;
    end
endmodule
```

a[2]	a[1]	a[0]	y[1]	y[0]
0	0	0	1	1
0	0	1	1	1
0	1	0	1	1
0	1	1	1	1
1	0	0	0	0
1	0	1	0	0
1	1	0	0	0
1	1	1	0	0

"don't cares" are also acceptable:

a[2]	a[1]	a[0]	y[1]	y[0]
0	X	X	1	1
1	X	X	0	0

7. (5 points) How many clock cycles of delay will the code below impose between input "inval" and output "c"?

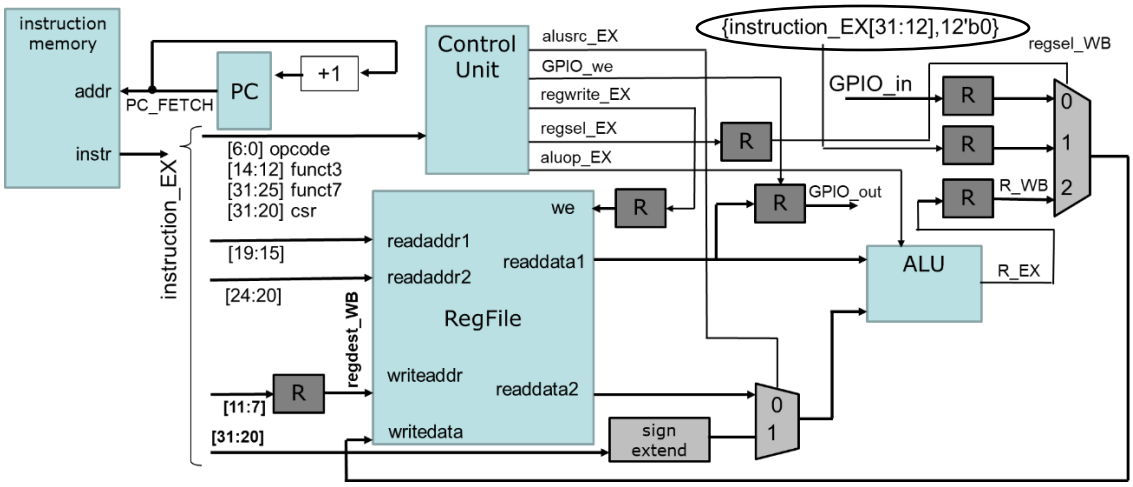
```
always_ff @(posedge clk) begin
  a <= inval;
  b <= a;
  c <= b;
  d <= c;
end
```

3 cycles

Points: _____

Part 3: Pipelined Microarchitecture

Consider the design of the 3-stage pipelined CPU design shown below:



a. (10 points) What is the purpose of the circled portion of the design?

Routes the upper immediate field of the U-type instruction (LUI) into a mux that can potentially select it for writing into the register file.

b. (5 points) What is the purpose of the 'GPIO_we' signal?

Updates the 'GPIO_out' signal.

c. (20 points) In the following table, mark the stage holding each of the instructions listed in the first three rows. Use the following abbreviations: fetch = "F", decode/execute = "E", write back = "W". Assume the instructions appear consecutively and the first instruction is fetched in cycle 0. Give the state of each of the control signals shown in the last six rows in each of the 6 cycles. Use "U" to denote an unknown value and "X" for don't care.

	cycle 0	cycle 1	cycle 2	cycle 3	cycle 4
`addi x2,x2,1`	F	E	W		
`mul x3,x3,x4`		F	E	W	
`add x3,x3,x5`			F	E	W
regwrite_EX		1'b1	1'b1	1'b1	
regwrite_WB			1'b1	1'b1	1'b1
GPIO_we		1'b0	1'b0	1'b0	
regsel_EX		2'd2	2'd2	2'd2	
regsel_WB			2'd2	2'd2	2'd2
alusrc_EX		1'b1	1'b0	1'b0	

d. (5 points) Does this CPU support branch and jump instructions? Why or why not?

No, the only value fed into the PC is PC+1.

Points: _____